

COMBINATORICS

CATALAN NUMBERS HOMEWORK

ROB TOMPKINS

Theorem 1. *Let C_n represent the Catalan numbers, with*

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

. We define the Catalan numbers as the number of valid ways to arrange n -pairs of left-right parentheses where every opening bracket has a subsequent closing bracket, and they are nested correctly. We want to show that C_n as defined above, also represents the number of binary trees with n vertices.

Proof. Our proof technique is the mechanics of reduction. We reduce one problem to another based on definition. Our goal will be to show that there is a bijective correspondence between the two definitions and their outcomes.

The fundamental idea behind the proof will be that every binary tree is determined by a root and a pair of smaller trees. So counting binary trees in this fashion is a split of counting two independent sub-Catalan objects.

Suppose Φ defines a map from binary trees to balanced parentheses strings. We do the following iteratively: (a) the empty tree maps to the empty string, and (b) if a tree has a root, a left subtree L , and a right subtree R , then we write $\Phi(T) = (\Phi(L))\Phi(R)$. $\square$